Paths in the unit type: for any x, y in the unit type the type x = y is equivalent to the unit type

The Univalent Foundations Program

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Editorial note. The counted statement is the author theorem labelled thm:path-unit (source lines 1540-1542); the proof span 1548-1559 is the author's proof environment, retained line for line.

Editorial note. The statement and proof cite no external labels, so no external references are declared.

Editorial scope. One original result of the HoTT book chapter cha:basics, counted once: the contractibility of path types in the unit type up to equivalence. Only the counted theorem and its author proof are retained; the author's warning about the tempting-but-failing naive strategy is commentary outside the counted statement and proof.

Allowed standard prerequisites (editorial).

- The unit type with its induction principle: every element is definitionally the canonical element.
- Equivalences via quasi-inverse: a function with a two-sided inverse up to identification.
- The induction principle for identity types (path induction).

Theorem 0.0.1. *For any $x, y : \mathbf{1}$, we have $(x = y) \simeq \mathbf{1}$.*

Proof. A function $(x = y) \to \mathbf{1}$ is easy to define by sending everything to $\star$. Conversely, for any $x, y : \mathbf{1}$ we may assume by induction that $x \equiv \star \equiv y$. In this case we have $\mathsf{refl}_\star : x = y$, yielding a constant function $\mathbf{1} \to (x = y)$.

To show that these are inverses, consider first an element $u : \mathbf{1}$. We may assume that $u \equiv \star$, but this is also the result of the composite $\mathbf{1} \to (x = y) \to \mathbf{1}$.

On the other hand, suppose given $p : x = y$. By path induction, we may assume $x \equiv y$ and p is refl_x. We may then assume that x is $\star$, in which case the composite $(x = y) \to \mathbf{1} \to (x = y)$ takes p to refl_x, i.e. to p. □